Chaitanya Bharathi Institute of Technology (CBIT)

Department of Computer Science and Engineering

CASE STUDY 5: WINE QUALITY PREDICTION

Statistical Analysis and Multiple Linear Regression on Physicochemical Properties

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1. Problem Statement and Objectives

Wine quality assessment traditionally relies on certified human sensory panels, which is time-intensive, expensive, and subject to taster fatigue and subjectivity. The objective of this case study is to investigate whether physicochemical laboratory measurements can predict wine quality scores using statistical tests and multiple linear regression.

Key goals include: (1) conducting descriptive and inferential statistical evaluations for core chemical markers (alcohol, volatile acidity, residual sugar), (2) evaluating dispersion and testing the significance of linear relationships via hypothesis testing, (3) exploring the feature space and checking for collinearity, (4) training a multiple linear regression model on an 80:20 train-test split, and (5) providing evidence-backed business recommendations regarding the feasibility of replacing or augmenting human sensory panels.

2. Dataset Description

The analysis is conducted on the red wine variant of the Portuguese 'Vinho Verde' dataset compiled by Cortez et al. (2009). The red wine dataset contains 1,599 observations across 11 continuous physicochemical predictor variables and 1 integer quality target (median score assigned by at least three sensory experts on a 0 to 10 scale). The red wine subset was selected as the primary dataset because mixing red and white wines would require an artificial dummy variable and conflate two distinct chemical profiles.

Data hygiene checks revealed 0 missing values and 240 duplicate rows (15.01%). These duplicate rows were dropped prior to modelling to prevent data leakage across train/test splits and guard against artificially optimistic evaluation metrics. The resulting modelling dataset consists of 1,359 clean, unique wine samples.

3. Methodology

The methodological workflow integrates rigorous statistical testing with supervised predictive modelling. First, descriptive statistics (mean, median, mode, std, skewness) and coefficient of variation (CV) were calculated. A formal two-tailed Pearson correlation test was executed with alpha = 0.05. For regression modelling, the dataset was split 80:20 (1,087 train samples, 272 test samples) with random_state=42. To prevent data leakage, StandardScaler was fitted strictly on training features and subsequently used to transform the test set.

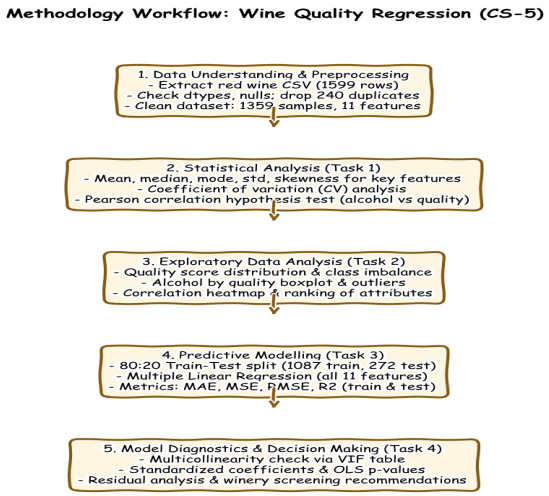


Figure 1: End-to-end wine quality regression and diagnostic workflow

Multicollinearity was diagnosed prior to coefficient interpretation using Variance Inflation Factor (VIF) and statsmodels OLS p-values. Model evaluation was conducted on unseen test data using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R2).

4. Statistical Findings & Exploratory Data Analysis

Task 1a & 1b: Central Tendency, Dispersion, and Relative Variability

Descriptive statistics were computed for alcohol, volatile acidity, and residual sugar (Table 1). Alcohol exhibited moderate right-skewness (+0.86) with mean 10.43% and median 10.20%. Volatile acidity showed right-skewness (+0.73) with mean 0.53 g/dm3 and median 0.52. Residual sugar displayed severe positive skewness (+4.55), reflecting that the majority of Vinho Verde wines are dry (median 2.20 g/dm3, mode 2.00 g/dm3), with a sparse tail of sweet dessert-style outliers.

Table 1: Descriptive Statistics and Skewness (Task 1a)

Attribute	Mean	Median	Mode	Std Dev	Skewness
alcohol	10.4323	10.20	9.50	1.0821	0.8598
volatile acidity	0.5295	0.52	0.50	0.1830	0.7293
residual sugar	2.5234	2.20	2.00	1.3523	4.5482

Coefficient of variation (CV) analysis revealed: Alcohol CV = 10.37%, Volatile Acidity CV = 34.57%, and Residual Sugar CV = 53.59%. Residual sugar exhibits the greatest relative variability. CV is the appropriate comparative measure because standard deviation is scale-dependent, whereas CV normalizes dispersion by the mean, permitting valid comparison across differing physical units (% vol vs g/dm3).

Task 1c: Pearson Correlation Hypothesis Test

A two-tailed hypothesis test was conducted on alcohol vs quality (H0: r = 0 vs H1: r != 0). The Pearson correlation was r = 0.4803 with p-value = 2.28e-79. Since p < 0.05, we reject H0 and conclude that a statistically significant, moderate positive linear relationship exists.

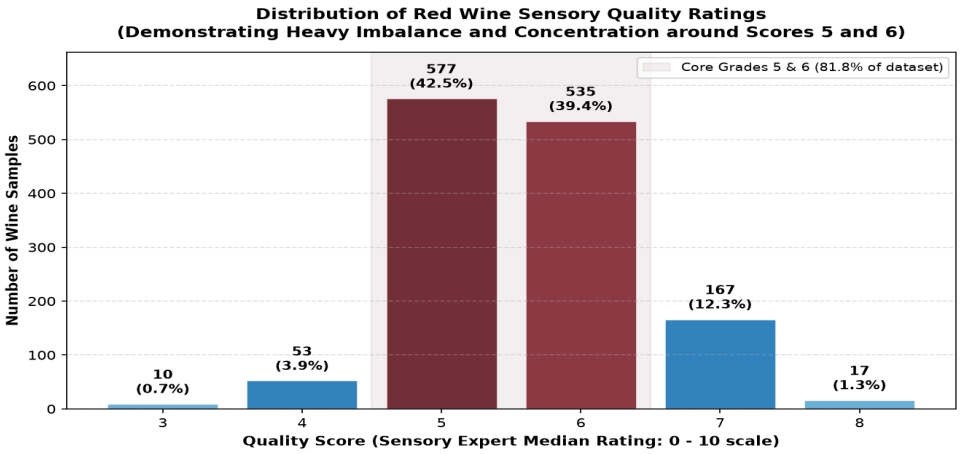


Figure 2: Wine quality score distribution showing severe concentration around scores 5 and 6 (81.8% of wines)

Task 2b: Alcohol Distribution Grouped by Quality Score

Boxplots illustrate a systematic upward trend in median alcohol content across quality grades: median alcohol increases steadily from 9.60% at score 5 to 10.50% at score 6, 11.60% at score 7, and 12.50% at score 8. A series of high-alcohol outliers appears in mid-tier wines (reaching up to 14.9%), indicating that high alcohol is necessary but not sufficient on its own to achieve superior sensory scores.

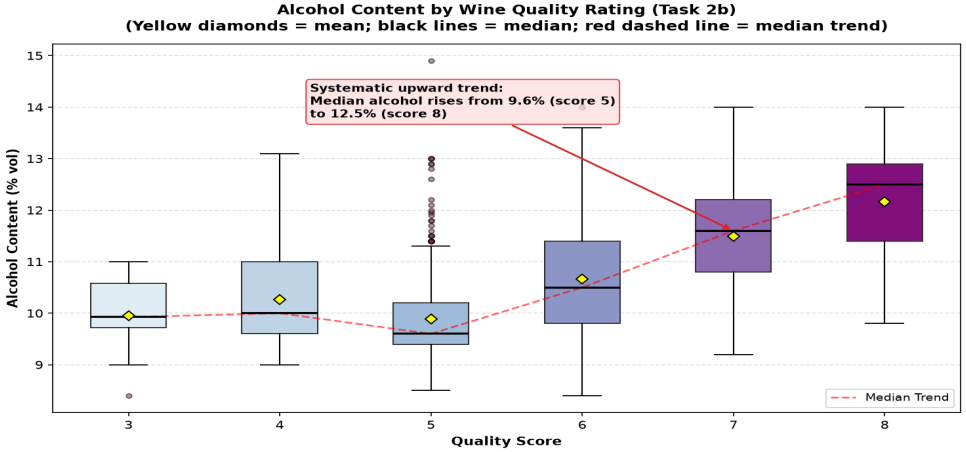


Figure 3: Alcohol content distribution across quality tiers with median trend and outliers

Task 2c & 2d: Correlation Heatmap and Predictor Ranking

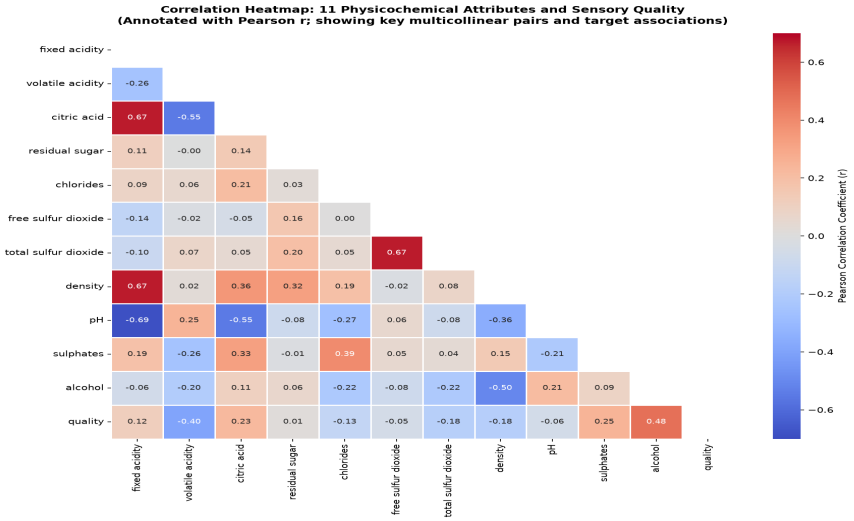


Figure 4: Correlation heatmap of all 11 physicochemical attributes and quality

Correlation ranking (Task 2d) identified the top two quality predictors: (1) Alcohol (r = +0.4803, moderate positive), and (2) Volatile Acidity (r = -0.3952, moderate negative, reflecting vinegar-like acetic acid spoilage). Noticeable inter-feature correlations include fixed acidity with citric acid (r = +0.67), fixed acidity with density (r = +0.67), and fixed acidity with pH (r = -0.68), signalling potential collinearity in multiple regression.

5. Predictive Modelling and Model Diagnostics

A multiple linear regression model was trained on all 11 physicochemical features with an 80:20 train-test split (1,087 train, 272 test). Test-set performance yielded MAE = 0.5041 quality points, MSE = 0.4310, and RMSE = 0.6565 quality points. The test R2 of 0.3915 aligns closely with train R2 (0.3520), verifying that the model generalizes stably without overfitting.

Table 2: Multiple Linear Regression Performance Metrics (Task 3c)

Metric	Value	Interpretation
MAE	0.5041	Average absolute error in quality points
MSE	0.431	Average squared error (penalises large errors more)
RMSE	0.6565	Root MSE in quality points (same unit as target)
R2 (test)	0.3915	Fraction of quality variance explained by the model
R2 (train)	0.352	Train R2 for overfitting check

Multicollinearity Diagnosis (VIF Analysis):
Variance Inflation Factor (VIF) evaluation on training data identified moderate-to-high multicollinearity: Fixed Acidity exhibited VIF = 7.95, and Density had VIF = 6.31 (both VIF > 5). This collinearity arises because fixed acidity directly influences solution density and pH (r = -0.68). While overall predictions remain unbiased, individual partial regression slope coefficients for these variables must be interpreted cautiously.

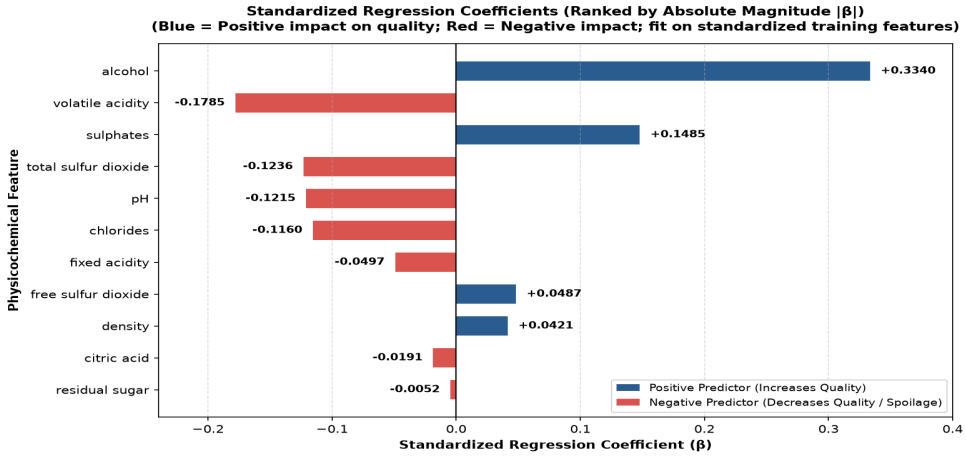


Figure 5: Standardized regression coefficients ranked by absolute magnitude with direction coding

Standardized coefficients rank relative predictor impact: 1. Alcohol (beta = +0.3340), 2. Volatile Acidity (beta = -0.1785), 3. Sulphates (beta = +0.1485), 4. Total Sulfur Dioxide (beta = -0.1236), and 5. pH (beta = -0.1215). Statsmodels OLS confirmed that volatile acidity, chlorides, total sulfur dioxide, sulphates, and alcohol are statistically significant ($p < 0.05$).

Model Diagnostics and Residual Analysis (Task 3e)

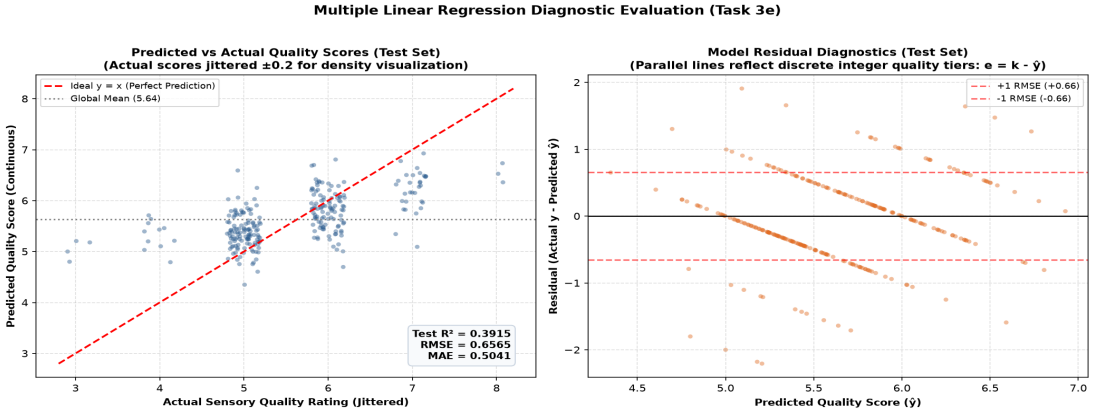


Figure 6: Predicted vs actual quality with jitter (left) and residual plot showing integer tier lines (right)

Predicted vs actual plots reveal that linear regression compresses predictions into a narrow band between 5.0 and 6.5. Because wine quality is a discrete ordinal target (3 to 8), continuous regression shrinks predictions toward the central training mean (5.63), underpredicting exceptional wines (score 7-8) and overpredicting poor wines (score 3-4). The residual plot exhibits distinct parallel downward lines (e = k - y_pred) corresponding to each discrete score integer level.

6. Business Decision-Making (Task 4)

Task 4a: Recommended Laboratory Screening Measurements

Based on the empirical evidence, the winery should implement automated quality screening on two key chemical markers:

1. **Alcohol Content:** Strongest positive correlation (r = +0.4803), largest standardized coefficient (+0.3340), and p < 0.0001. Higher alcohol corresponds to complete grape maturation, sound sugar fermentation, and desirable body.
2. **Volatile Acidity:** Second strongest correlation (r = -0.3952), largest negative standardized coefficient (-0.1785), and p < 0.0001. Elevated acetic acid indicates bacterial spoilage and poor sanitation.

Task 4b: Model Reliability for Sensory Panel Replacement

The model is NOT reliable enough to replace expert sensory panels. With an R2 of 0.3915 (explaining ~39% of variance) and an RMSE of 0.6565 quality points on a discrete scale, the model frequently misjudges wines by an entire rating tier. It should be deployed strictly as a pre-screening filter to detect flawed batches before submission to tasting panels.

Task 4c: Real-World Limitations

1. The dataset is restricted to red Vinho Verde wines from northern Portugal; chemical benchmarks do not generalize to other varietals or regions.
2. Target quality is an ordinal median score from expert tasters, which embodies sensory subjectivity and noise that linear regression cannot fully capture.

7. Conclusion and Limitations

This case study successfully characterized the relationship between physicochemical attributes and red wine quality. Multiple linear regression achieved moderate predictive utility (RMSE = 0.66, R2 = 0.39), identifying alcohol and volatile acidity as the primary drivers of sensory preference. While valuable for rapid automated vat screening, sensory tasting panels remain indispensable.

Key limitations include: (1) restriction to red Vinho Verde wines, (2) ordinal nature of sensory scores, (3) severe class imbalance (scores 5 and 6 dominate), (4) inability of linear models to capture non-linear biochemical interactions, and (5) absence of grape varietal, vintage, or brand metadata.

8. References & Attribution

1. Cortez, P., Cerdeira, A., Almeida, F., Matos, T., and Reis, J. (2009). Modeling wine preferences by data mining from physicochemical properties. *Decision Support Systems*, Elsevier, 47(4):547-553.
2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
3. Seabold, S., and Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*.
4. GitHub Repository: https://github.com/katthi-goutham-reddy/FDS-case-study

Attribution: Code structure drafted with AI assistance (Claude); executed, verified and interpreted by Katthi Goutham Reddy. Dataset source: Cortez et al. (2009), UCI Machine Learning Repository.